

HABIB IBRAHIM TOURE

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Computer Science Co-op Student @ uOttawa — AI/ML Systems & Applied Engineering

EDUCATION

University of Ottawa

Ottawa, ON

B.Sc. (Specialized) in Computer Science, Co-op | International Student Merit Scholarship – \$36,000/yr Sept. 2024 – 2028

- Relevant coursework: Data Structures and Algorithms, Computer Architecture, Introduction to Software Engineering

EXPERIENCE

Data Annotation & AI Training Specialist

June 2025 – Oct. 2025

DataAnnotation

Remote, ON

- Built Python automation scripts to streamline repetitive ML training-data annotation workflows, reducing manual effort by ~30% across 5+ ML projects
- Evaluated model outputs and delivered structured feedback that measurably improved labeling consistency and model accuracy
- Identified systematic annotation errors in collaboration with a remote AI team, directly refining training data quality guidelines

PROJECTS

AITradingAgent | *Python, Redis, PostgreSQL/TimescaleDB, Pandas, NumPy, Docker*

June 2026 – Present

- Designed a rigorous statistical validation pipeline (train/test split, significance thresholds, outlier-sensitivity diagnostics) to evaluate trading strategies before any deployment
- Validated a statistically significant trading edge on multi-year historical data (~700K+ OHLCV bars) with modeled slippage and fees, passing simulated prop-firm risk-evaluation criteria
- Rejected 3 underperforming strategies based on quantitative evidence rather than intuition, each backed by a full diagnostic report
- Currently validating performance in live paper trading while extending the system with ML-based parameter optimization

MARA | *Python, Claude API, Whisper, CUDA, PyQt5, Selenium*

Apr. 2026 – Present

- Architected a local AI voice assistant using Claude Sonnet's native tool-calling for autonomous routing between conversation, system actions, and specialized tools, achieving sub-second response latency
- Built a hybrid on-device/cloud ML pipeline: CUDA-accelerated Whisper for real-time speech-to-text combined with LLM-based reasoning and vision for on-screen code analysis
- Engineered persistent, encrypted cross-session memory with automatic background extraction, enabling context-aware personalization without explicit re-prompting

Trading Model Explainability Platform | *Python, SHAP, hmmlearn, FastAPI, gVisor*

Aug. 2026 – Present

- Building a model-agnostic explainability tool using SHAP (TreeExplainer/KernelExplainer) to attribute individual trading decisions to input features, with mandatory additivity and synthetic-model sanity tests
- Designing a 3-state Gaussian Hidden Markov Model for market regime detection using filtered (causal-only) probabilities to prevent lookahead bias
- Architecting sandboxed (gVisor) execution for untrusted user-submitted models, enforcing network isolation and strict resource/time limits

TECHNICAL SKILLS

Languages: Python, SQL, TypeScript, JavaScript, Java

ML / Data: PyTorch, Scikit-learn, NumPy, Pandas, SHAP, hmmlearn, Pandera, CUDA / GPU Inference

AI / LLM: Anthropic Claude API, Native Tool-Calling, Whisper, Prompt Engineering

Infrastructure: PostgreSQL, Redis, Docker, FastAPI, REST APIs

Tools: Git, Linux, CI/CD, Agile

Other: Bilingual (English/French)